

CreditRisk AI — Solution Approach

Defaulter Risk Management & Visual Telemetry

1. Executive Summary

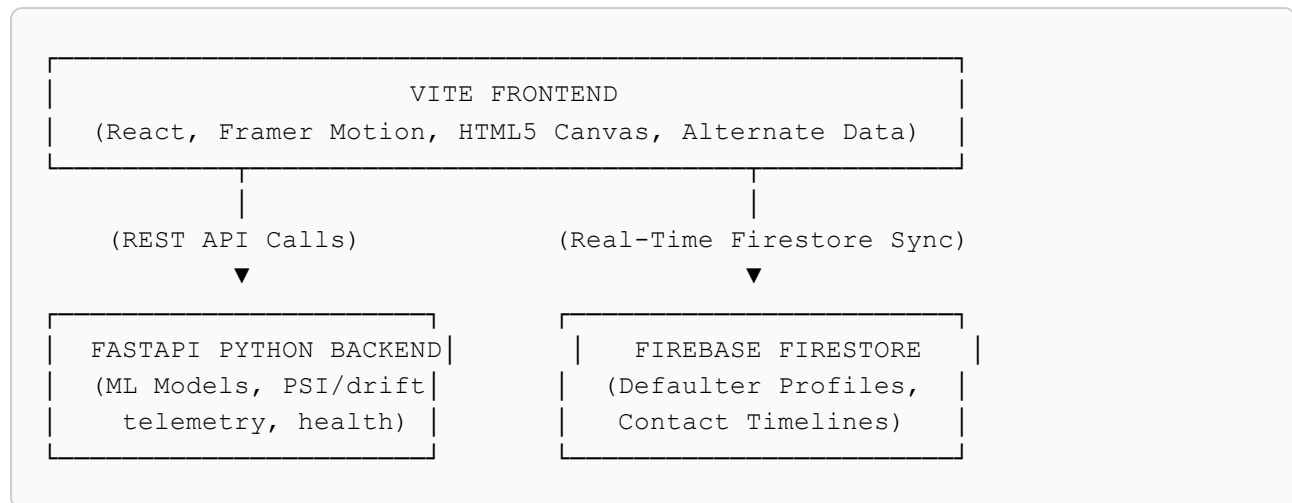
CreditRisk AI is a dynamic web application designed to optimize portfolio risk management for financial institutions. By consolidating alternate data streams, geospatial clusters, and persistent contact histories into a single reactive interface, the platform empowers risk officers to take quick and informed actions against NPAs (non-performing assets).

2. Problem Statement

Standard risk management workflows face several challenges:

- **Static & Delayed Data:** Risk histories are locked in static spreadsheets, causing delayed actions.
- **Interface Lockups:** Network latency on database writes freezes the UI, decreasing worker efficiency.
- **Lack of Spatial Context:** Visualizing risk density geographically is difficult with standard tables.

3. Proposed Architecture



The system comprises three core layers: a React client layer for dynamic telemetry, a serverless Firebase layer for persistent database syncing, and a Python FastAPI backend for statistical metrics.

4. Core Solutions Implemented

Real-Time Defaulter Intelligence Portal

- **Asynchronous, Non-Blocking Saves:** Contact logging form submissions trigger Firestore writes in the background. The user interface updates instantly, bypassing network latency.
- **Optimistic Deduplication:** Uses client-side generated UUIDs to resolve latency compensation, merging local states and remote updates with zero duplication.
- **Named Collections:** Automatically records documents under the borrower's name in Firestore, enhancing database readability.

3D Defaulter Risk Globe

- **Interactive Geometry:** Renders a fully responsive 3D globe using a high-performance Canvas 2D engine.
- **Risk Contagion Bezier Arcs:** Selecting a city cluster draws curved Bezier lines with flowing particles to visualize risk contagion paths.

- **Search & Focus:** Users can search for cities, rotating the globe automatically to bring the selected point to the front.
- **Visual Skins:** Offers live switches between three visual skins (Blueprint, Magma, and Matrix).

5. Security & Configuration

- **Environment Variables:** Real Firebase credentials are loaded via Vite-prefixed variables from a local `.env` file.
- **Git Safety:** The `.env` file is hidden using `.gitignore`, ensuring sensitive API keys are never leaked to public repositories.